

Forces and Torque on a Disk in the Stokes Regime

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1 Forces and Torque

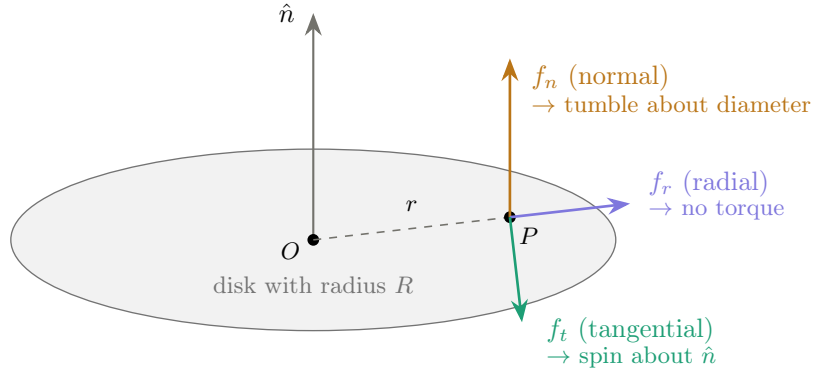


Figure 1: A force $\vec{f}$ is exerted against a disk centred at $\vec{O}$ with radius R at point P , $|\vec{O} - \vec{P}| = r$. That force can be decomposed into components normal to the disk, $f_n \hat{n}$, radial to the disk, $f_r \hat{s}$, and tangential to the disk $f_t \hat{t}$.

Let $\vec{O}$ be the centre of a disk with radius R . This disk sits in cytoplasm with some viscosity μ large enough to be in the stokes regime.

A microtubule (MT) impinges on this disk at some point $\vec{P}$ such that $|\vec{O} - \vec{P}| = r$ and exerts a force $\vec{f}$ on the disk.

We can decompose $\vec{f}$ into the components, extending from $\vec{m}_i$, $f_r \hat{s}$ radial, $f_n \hat{n}$ normal, and $f_t \hat{t}$ tangential ($\hat{t} = \hat{n} \times \hat{s}$). Then

$$\vec{f} = f_r \hat{s} + f_t \hat{t} + f_n \hat{n} \quad (1)$$

and the torque is simple to calculate because the radial component of force exerts no torque

$$\vec{\tau} = (r \cdot \hat{s}) \times \vec{f} = (r \cdot f_t) \hat{n} + (r \cdot f_n) \hat{t} \quad (2)$$

From the symmetry of a disk, we know there are four modes of motion, each with their own stokes resistance coefficient:

1. $\zeta_{\parallel} = 16\mu R$ resists radial motion in plane with the disk,
2. $\zeta_{\perp} = \frac{32}{3}\mu R$ resists broadside motion out of plane with the disk,
3. $\zeta_{\text{spin}} = \frac{32}{3}\mu R^3$ resists spin about the centre of the disk,
4. $\zeta_{\text{tumble}} = \frac{32}{3}\mu R^3$ resists tumbling about the diameter of the disk,

Set $\zeta_\omega = \zeta_{\text{spin}} = \zeta_{\text{tumble}} = \frac{32}{3}\mu R^3$. It is unintuitive and surprising that the drag is the same for spinning and tumbling. These results are explicitly stated in [1].

The equations of motion for a disk experiencing force from a single MT then are

$$\vec{U} = \frac{f_n}{\zeta_\perp} \hat{n} + \frac{1}{\zeta_\parallel} (f_r \hat{s} + f_t \hat{t}) \quad (3)$$

$$\vec{\omega} = \frac{r}{\zeta_\omega} (f_t \hat{n} - f_n \hat{t}) \quad (4)$$

with $\vec{P}$ following along as with a normal rigid body.

When there are many MTs, we have a collection of points exerting force $\{\vec{P}_i\}$, each with their own $r_i, \hat{n}_i, \hat{t}_i, \hat{s}_i$. Since the system is linear, the equations of motion for the collective system are

$$\vec{U} = \frac{1}{\zeta_\perp} \left(\sum_i f_{n,i} \cdot \hat{n}_i \right) + \frac{1}{\zeta_\parallel} \left(\sum_i f_{r,i} \cdot \hat{s}_i + f_{t,i} \cdot \hat{t}_i \right) \quad (5)$$

$$\vec{\omega} = \frac{1}{\zeta_\omega} \left(\sum_i r_i (f_{t,i} \cdot \hat{n}_i - f_{n,i} \cdot \hat{t}_i) \right) \quad (6)$$

2 impingement on disk test

Let $\vec{R}_M$ be the position of an MTOC nucleating an MT with direction $\hat{m}$. There is a metaphase plate with a centre of mass at $\vec{R}_D$, unit normal $\hat{n}$ and radius a .

Claim:

the MT hits the disk when

$$t^* = -\frac{\vec{R} \cdot \hat{n}}{\hat{m} \cdot \hat{n}} > 0 \text{ and } |\vec{R}_M + t^* \hat{m}| \leq a. \quad (7)$$

Proof:

The MT leaves from $\vec{R}_M$ and travels like a ray

$$\vec{x}(t) = \vec{R}_M + t\hat{m}, \quad t \geq 0 \quad (8)$$

The position of the MTOC relative to the centre of the disk is $\vec{R} = \vec{R}_M - \vec{R}_D$. When the MT's ray hits the plane orthogonal to the disk's $\hat{n}$, the following must be true

$$(\vec{R} + t\hat{m}) \cdot \hat{n} = 0 \quad (9)$$

$$\implies \vec{R} \cdot \hat{n} + t(\hat{m} \cdot \hat{n}) = 0 \quad (10)$$

$$\implies t^* = -\frac{\vec{R} \cdot \hat{n}}{\hat{m} \cdot \hat{n}} \quad (11)$$

This is the length of the MT when it crosses the plane.

- If $t^* = 0$, the MT is moving parallel to the plane and never intersects it,
- if $t^* < 0$, the MT is moving away from the plane and never intersects it,
- if $t^* > 0$, the MT is moving towards the plane and intersects it when at length t^* .

The point on the plane where the MT hits is $P = \vec{R}_M + t^* \hat{m}$, and the MT hits the disk when $|P| \leq a$.

So the MT hits the disk when

$$t^* = -\frac{\vec{R} \cdot \hat{n}}{\hat{m} \cdot \hat{n}} > 0 \text{ and } |\vec{R}_M + t^* \hat{m}| \leq a. \quad (12)$$

References

- [1] Sangtae Kim and Seppo J Karrila. *Microhydrodynamics: principles and selected applications*. Butterworth-Heinemann, 2013.